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Please allow this document to serve as Ronawk's response to the Request for Information by the Office of Science and Technology Policy (OSTP), the NITRD NCO for input from interested parties on the Development of an Artificial Intelligence (AI) Action Plan ("Plan").

The progress of AI technology and its numerous benefits seems inescapable, propelled largely by formidable market forces. It is crucial to delve into what a world with fully developed AI might look like. America is striving for a mutually beneficial result that improves life for everyone. While AI has numerous positive applications, this response will focus specifically on its impact on biotechnology.

Next-generation biotech will transform the American economy with groundbreaking solutions in healthcare and pharmaceuticals, agriculture and food production, energy and environmental sustainability, and manufacturing and material sciences, thereby providing national and economic security. The biotechnology landscape is undergoing a dramatic revolution, demanding immediate action. Legacy biomanufacturing is severely constrained by conventional frameworks rooted in chemistry, physics, mechanics, and energy. **To unlock the full potential of next-generation biotech, it is imperative that a biological perspective be incorporated into future biomanufacturing infrastructure.** We must employ the essential principles of biology, leveraging the synergies, rather than the limitations, from established bio-manufacturing models rooted in chemistry, physics, mechanics, and energy to revolutionize the paradigm of bio-innovation. By bio-engineering advanced biological technologies, America will transform the evolution of the next-generation biomanufacturing ecosystem and accelerate the future of bio-based innovation.

Biology is the field where scientific progress holds the highest potential to directly and profoundly enhance human life. Over the past century, we have eradicated some of the most perilous afflictions (smallpox, polio, tuberculosis, and measles) yet many diseases persist and overcoming them will be a monumental achievement. However, current biological research—whether on humans, animals, or plants is expensive and hindered by issues of inconsistency and reproducibility. Additionally, while vast amounts of biological data exist, the challenge lies more in its quality than its quantity.

One of the key challenges with biological data is its inherent complexity. Scientists recognize the difficulty of isolating the individual components within this intricate system. Even greater challenges arise when attempting to intervene in a precise or predictable manner. There remains a widespread belief that AI is only useful in a limited capacity—often summarized as, "AI can analyze data more effectively, but it can't generate new data or enhance its quality." Therefore, what we are left with is "garbage in, garbage out".

By accelerating the entire research process, AI has the potential to accelerate significant breakthroughs in biology. To unlock its full potential, AI should be leveraged to perform, direct, and enhance nearly every aspect of biological research.¹

¹ Dario Amodei, *Machines of Loving Grace* (October 2024)



Historically, a handful of key discoveries each decade has shaped our fundamental understanding of biology and driven the development of groundbreaking medical treatments. With the integration of AI into next-generation biotechnology, the pace of discovery can accelerate tenfold, potentially compressing 50–100 years of biological progress into just 5–10 years. AI-driven biological science will reduce the need for extensive iteration in non-clinical and pre-clinical studies by creating more precise cell and animal models or simulations that more accurately predict human outcomes. This will be especially transformative for drug development, where faster iteration cycles are critical. With the development of biological AI, America can accomplish decades' worth of progress in biology and medicine in just a few years.

While new challenges—such as ensuring widespread access to emerging technologies—will arise, it is worth considering how profoundly the world will improve if AI transforms only the field of biotechnology.

Next-generation biotechnology will create new services and products, that provide opportunities to grow the American economy and workforce and is the path to bettering American lives. If the 19th century was the century of chemistry, and the 20th century was the century of physics, the 21st century is the century of biology. For this bio-revolution to become a reality there must be significant improvements made in the manufacturing (both economic and process) of the biological raw materials needed to accelerate the next-generation bioeconomy. Advancements in AI-enabled biological tools and automated laboratories that reduce or eliminate the need for extensive experimental validation will represent a major breakthrough in overcoming existing bottlenecks and barriers.

America is not the only country to have this idea. For some, there is a pressing sense of American decline, especially relative to a powerful and more assertive China (e.g., Deepseek vs Open AI). China has indicated biotechnology and biomanufacturing are a top national priority starting with the 13th Five Year Plan and continuing into the 14th Five Year Plan with an emphasis on biomedicine and health, agriculture, industrial biotechnology, and frontier biotechnology, which includes synthetic biology.

Biotechnology is advancing quickly and converging with other technologies (e.g., AI) in ways that open doors to historic innovation. However, our adversaries may look to use these emerging technologies in ways that could harm America. AI-enabled biological tools will strengthen biosecurity and reduce biological threats by improving prediction, detection, prevention, and response. This includes enhancing bio surveillance and rapidly advancing the development of medical countermeasures, such as vaccines and antibodies, to address both intentional biological threats and naturally occurring infectious disease outbreaks. The ability of AI to potentially accelerate countermeasure design is particularly vital in emergency responses to biological threats. We must endeavor to safeguard the American bioeconomy, as foreign adversaries and strategic competitors alike use legal and illegal means to acquire America's technologies and data, including biological data, and proprietary or precompetitive information, which threatens America's economic competitiveness and national security. Allowing China to outpace us will be a disaster for America and the free world.

Investment in next-generation national biomanufacturing infrastructure is not only a good policy but one of the best investments America can make overall as it sets a path of innovation that will shape America's future. Unlike most military hardware, biotechnology can be employed productively in peacetime, supporting critical scientific, government, and industrial functions. We spend billions replenishing American military armament every year,

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and unlike traditional military armament, biotechnology offers immediate and enormous benefit for critical civilian applications.

The U.S. defense budget is around ~\$850 billion and only a small fraction is required to build a national biomanufacturing infrastructure. **Next-generation biotechnology will allow America to do more with less.** Working with American firms, the Administration should direct development funds to invest in next-generation biotechnology to counter the growing influence of China. The Administration should then invest more in next-generation biotechnology research and development that will unlock American innovation. Investment in a dedicated defensive national bioeconomy is an investment in American reindustrialization and jobs, American security, and American resilience—and one America can't afford not to make.

Free markets have the power to drive an unstoppable wave of innovation in America. Developing and sustaining a next-generation national biomanufacturing infrastructure will not only strengthen critical biomanufacturing capabilities but also create thousands of well-paying jobs domestically. A robust American bioeconomy can support the construction of domestic biomanufacturing facilities by utilizing advance market commitments for domestically produced bio manufactured goods that align with both military and civilian needs.

America must possess the determination and courage to move forward on all of the above to build a comprehensive biomanufacturing ecosystem before a global adversary catches up. America must exploit its lead if we wish to keep it. **America should be the world's most innovative and agile force in next-generation biotechnology.** Building strategic biomanufacturing capabilities is both sensible defense and crucial industrial policy that is aligned with current national objectives.

In the long run, a powerful American bioeconomy means more stability and security. The Trump Administration, Congress, and the Department of Defense (DoD) must work together to develop and fortify this essential pillar of the American economy. Doing such will build America's resilience against supply disruptions, provide counter measures against an inevitable attack, bring critical biomanufacturing home, and solidify the U.S. as the leading authority in a next-generation biomanufacturing ecosystem, while granting the American people key levers to shape the next era of this foundational technology.

A country with this capability can outbuild, outthink, outmaneuver, and outlast its opponents.

Sincerely,
Ronawk, Inc.

Thomas W. Jantsch
President & COO



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